
Date – 29th April 2026

MIT Academy of Engineering , Alandi, Pune

BTech WC ESE Project report- Part 2

Design sheet Year 2025-26 Term 2

Q-1 Problem statement –

Design an OFDM transmitter and receiver system with a 2×2 MIMO configuration using Simulink. The system should implement QPSK modulation, use 48 subcarriers with an FFT size of 64 and a cyclic prefix of 16, and incorporate OSTBC encoding and decoding over a Rayleigh fading channel.

FFT Length - 256

Data Subcarriers - 192

OFDM Symbols - 5

Cyclic Prefix Length -32

Type of Modulation -QPSK

Group Number –B15

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[illegible]

Answer

System performance matrix are as follows –

Q-1 (a) Data Rate (Updated)

Given:

- Number of subcarriers = **192**
- Modulation = **QPSK** → **2 bits/symbol**
- FFT size = **256**
- Cyclic Prefix = **32**

$$T_{\text{symbol}} = T_{\text{FFT}} + T_{\text{CP}} = 256 + 32 = 288$$

Data per OFDM symbol:

$$\text{Data} = N_{\text{subcarriers}} \times \text{bits per symbol} = 192 \times 2 = 384 \text{ bits}$$

Data Rate (per OFDM symbol):

$$384 \text{ bits per OFDM symbol}$$

With MIMO (2×2):

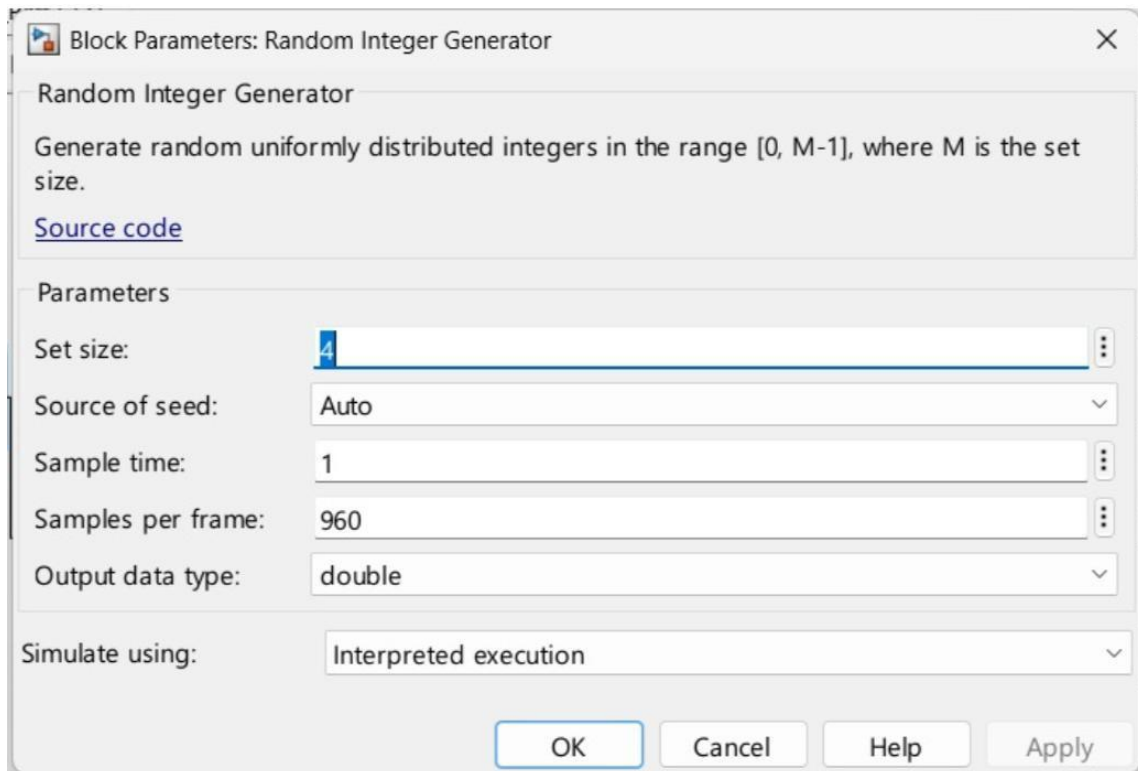
$$\text{Effective Data Rate} = 384 \times 2 = 768 \text{ bits/symbol}$$

(b) Quality Matrix

Parameter	Description	Observation
Bit Rate	Number of bits transmitted per second	High due to 192 subcarriers and further improved using MIMO (2× gain)
SNR	Signal-to-Noise Ratio	High SNR → System shows reliable performance
MSE	Mean Square Error	Low at high SNR (accurate signal recovery)
Correlation Coefficient	Similarity between transmitted & received signal	Close to 1 (indicates good system performance)
MOS	Mean Opinion Score (quality measure)	High at high SNR (good perceived quality)

Block Parameters:

Random Integer Block



Block Parameters: Random Integer Generator

Random Integer Generator

Generate random uniformly distributed integers in the range $[0, M-1]$, where M is the set size.

[Source code](#)

Parameters

Set size:

Source of seed:

Sample time:

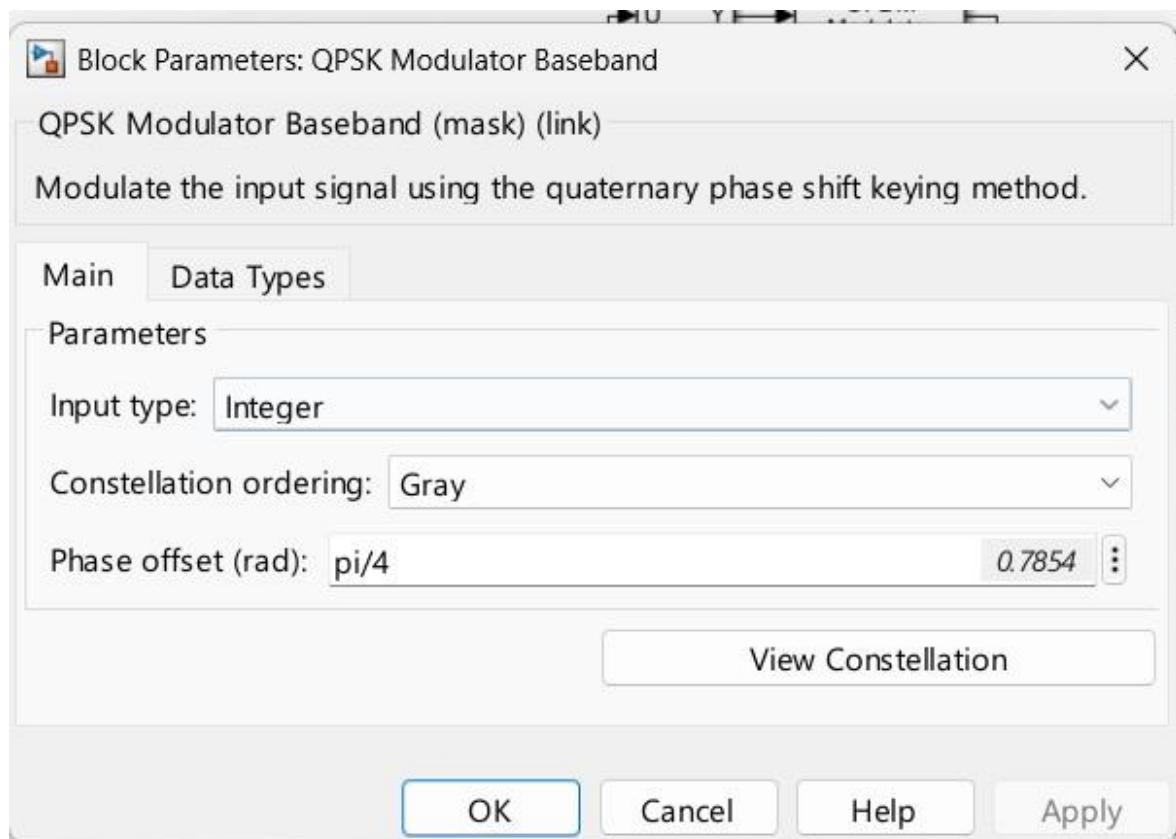
Samples per frame:

Output data type:

Simulate using:

OK Cancel Help Apply

QPSK Modulator



Block Parameters: QPSK Modulator Baseband

QPSK Modulator Baseband (mask) (link)

Modulate the input signal using the quaternary phase shift keying method.

Main Data Types

Parameters

Input type:

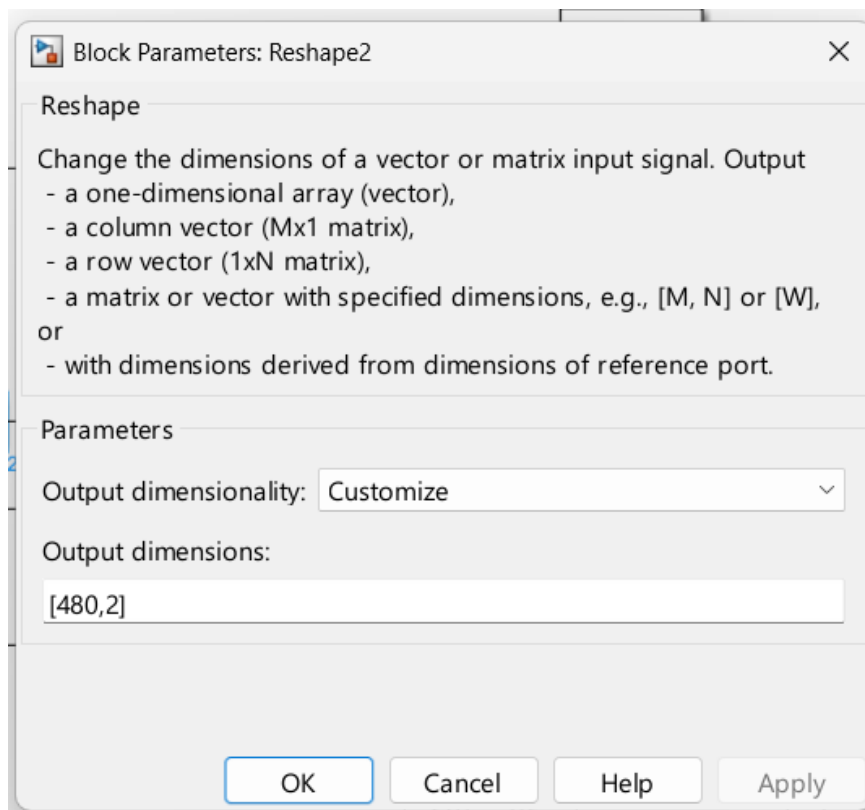
Constellation ordering:

Phase offset (rad): 0.7854

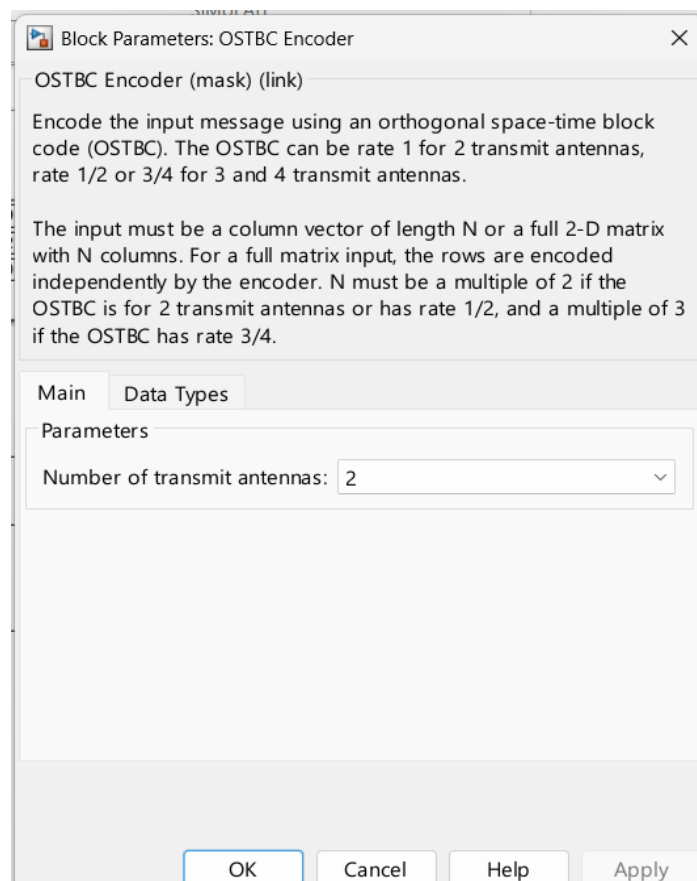
View Constellation

OK Cancel Help Apply

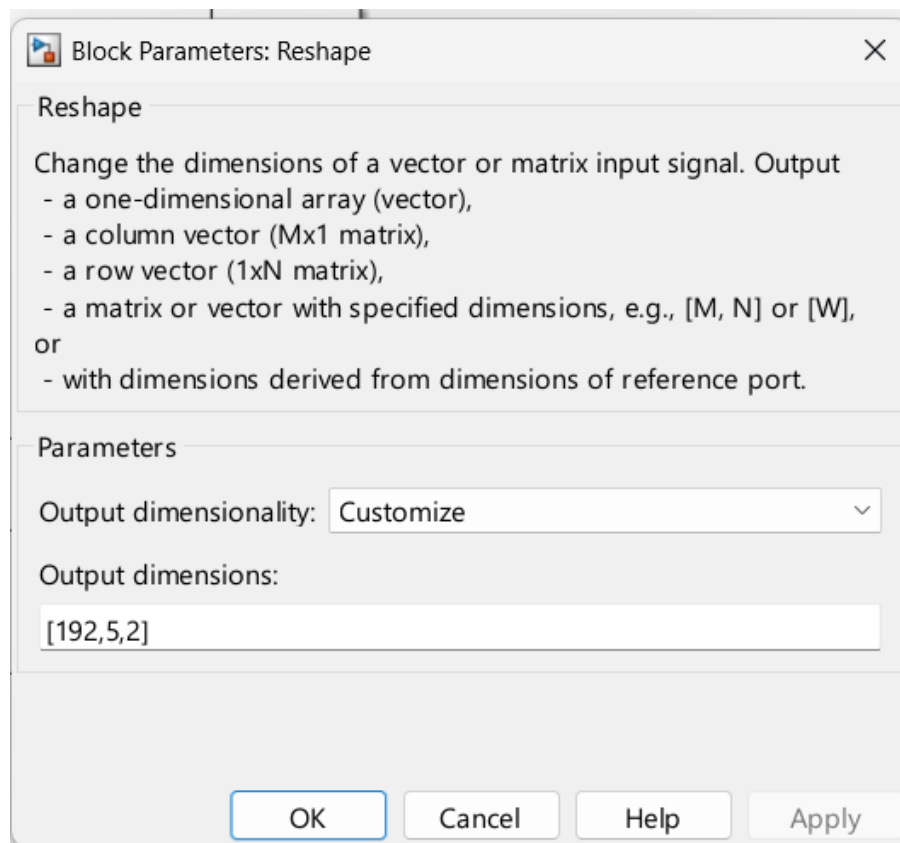
Reshape before OSTBC



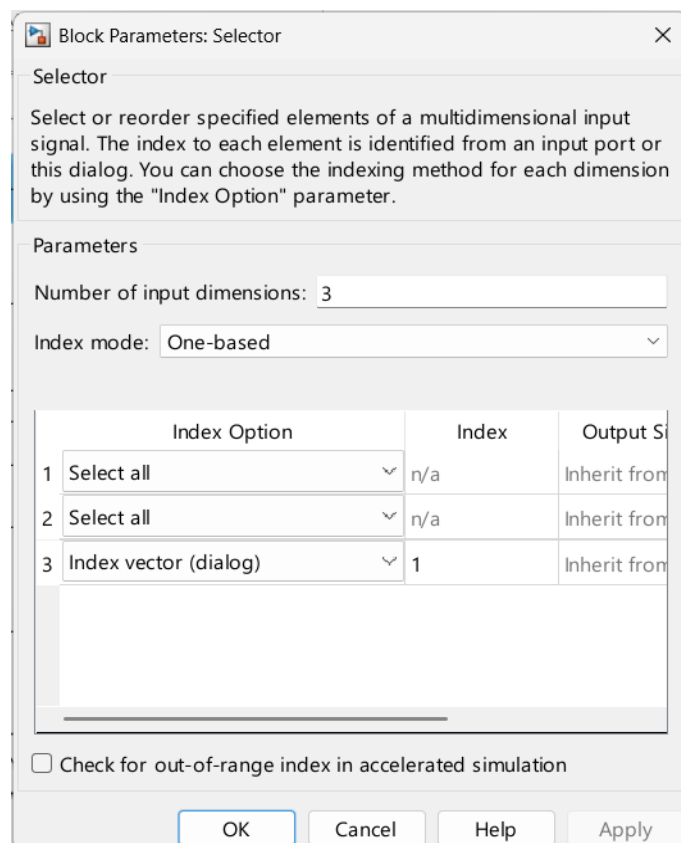
OSTBC Encoder



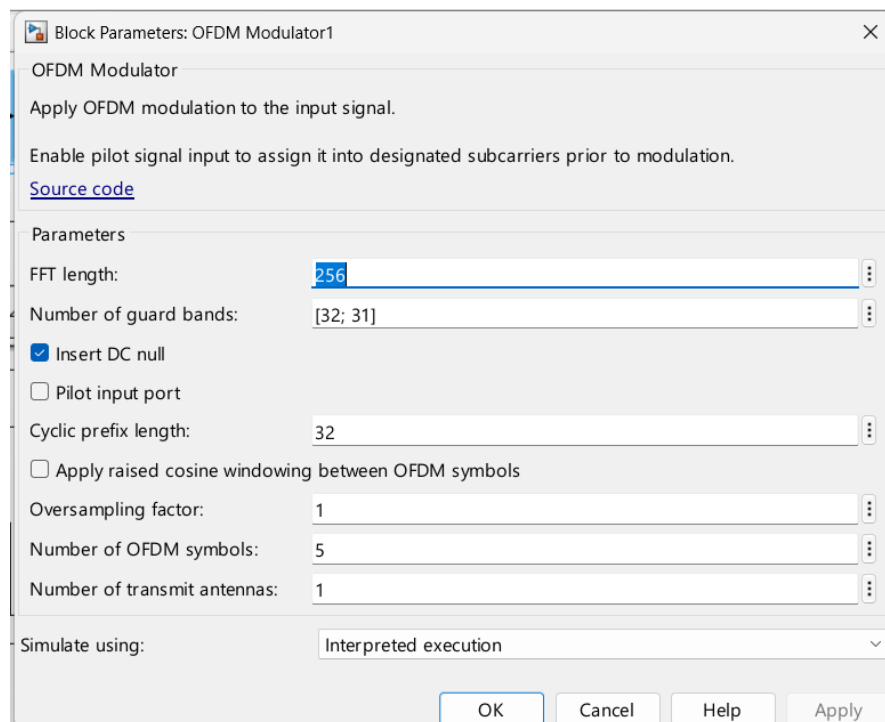
Reshape after OSTBC



Selector before OFDM Mod



OFDM Modulator



Block Parameters: OFDM Modulator1

OFDM Modulator

Apply OFDM modulation to the input signal.

Enable pilot signal input to assign it into designated subcarriers prior to modulation.

[Source code](#)

Parameters

FFT length: 256

Number of guard bands: [32; 31]

☒ Insert DC null

☐ Pilot input port

Cyclic prefix length: 32

☐ Apply raised cosine windowing between OFDM symbols

Oversampling factor: 1

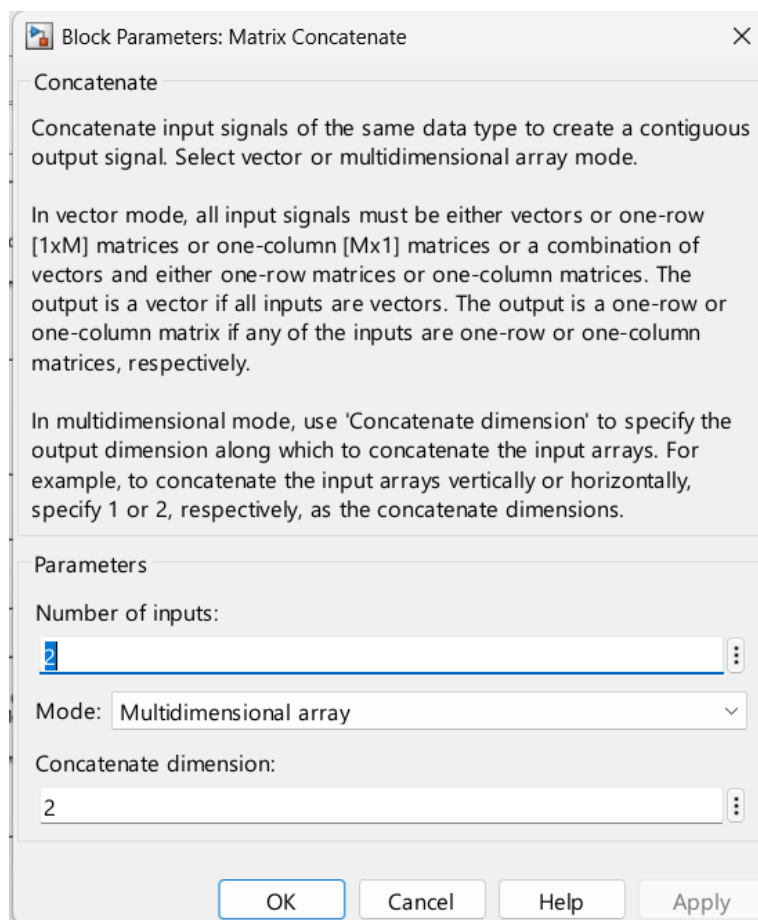
Number of OFDM symbols: 5

Number of transmit antennas: 1

Simulate using: Interpreted execution

OK Cancel Help Apply

Matrix Concatenate after OFDM Modulator



Block Parameters: Matrix Concatenate

Concatenate

Concatenate input signals of the same data type to create a contiguous output signal. Select vector or multidimensional array mode.

In vector mode, all input signals must be either vectors or one-row [1xM] matrices or one-column [Mx1] matrices or a combination of vectors and either one-row matrices or one-column matrices. The output is a vector if all inputs are vectors. The output is a one-row or one-column matrix if any of the inputs are one-row or one-column matrices, respectively.

In multidimensional mode, use 'Concatenate dimension' to specify the output dimension along which to concatenate the input arrays. For example, to concatenate the input arrays vertically or horizontally, specify 1 or 2, respectively, as the concatenate dimensions.

Parameters

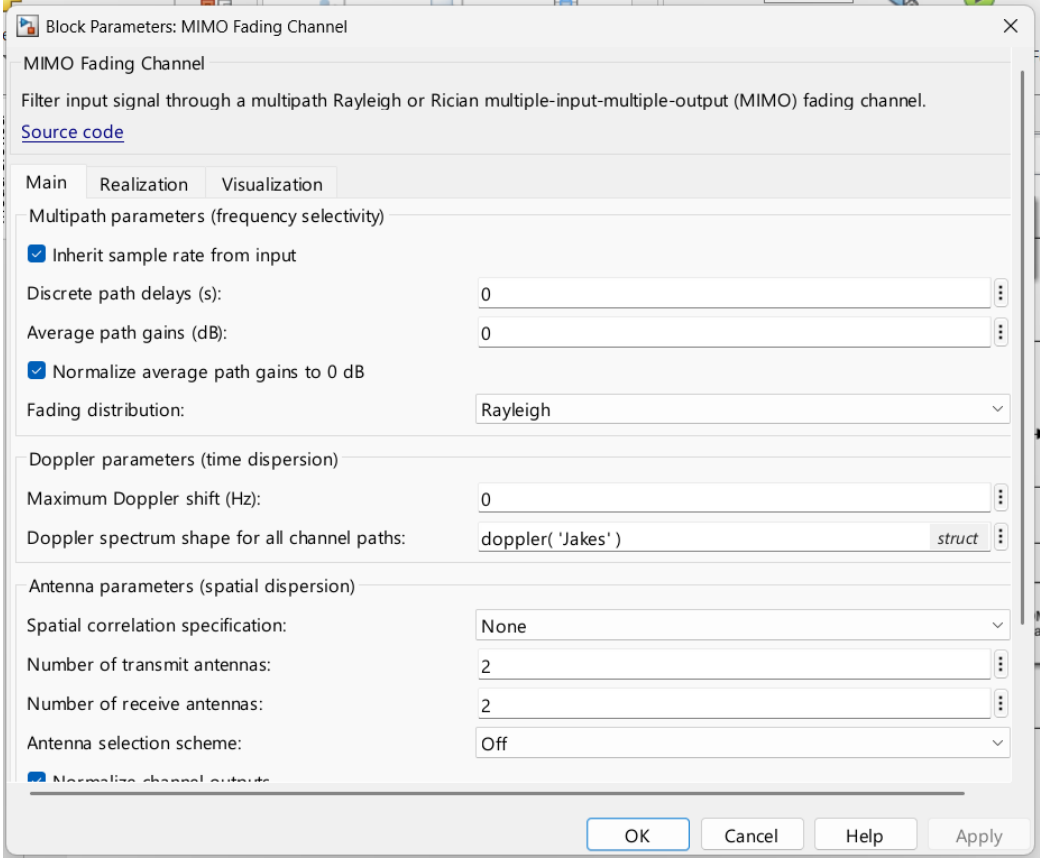
Number of inputs: 2

Mode: Multidimensional array

Concatenate dimension: 2

OK Cancel Help Apply

Rayleigh MIMO



Block Parameters: MIMO Fading Channel

MIMO Fading Channel

Filter input signal through a multipath Rayleigh or Rician multiple-input-multiple-output (MIMO) fading channel.

[Source code](#)

Main Realization Visualization

Multipath parameters (frequency selectivity)

☒ Inherit sample rate from input

Discrete path delays (s): 0

Average path gains (dB): 0

☒ Normalize average path gains to 0 dB

Fading distribution: Rayleigh

Doppler parameters (time dispersion)

Maximum Doppler shift (Hz): 0

Doppler spectrum shape for all channel paths: doppler('Jakes')

Antenna parameters (spatial dispersion)

Spatial correlation specification: None

Number of transmit antennas: 2

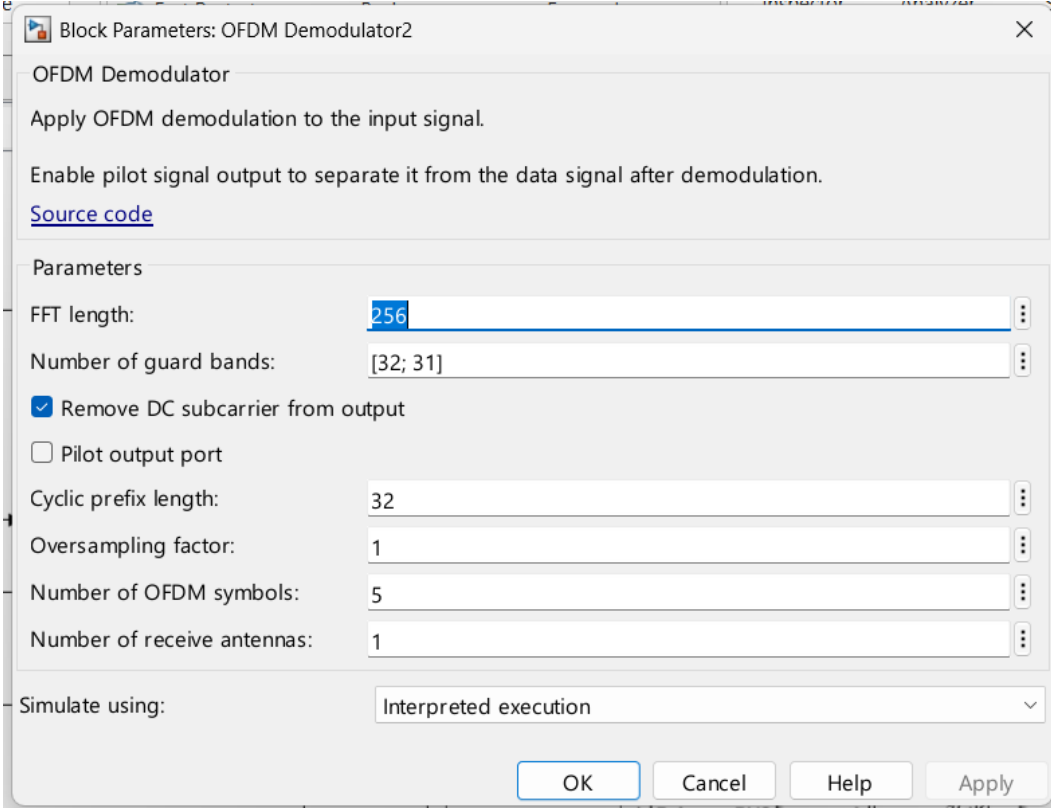
Number of receive antennas: 2

Antenna selection scheme: Off

☒ Normalize channel outputs

OK Cancel Help Apply

OFDM Demodulator



Block Parameters: OFDM Demodulator2

OFDM Demodulator

Apply OFDM demodulation to the input signal.

Enable pilot signal output to separate it from the data signal after demodulation.

[Source code](#)

Parameters

FFT length: 256

Number of guard bands: [32; 31]

☒ Remove DC subcarrier from output

☐ Pilot output port

Cyclic prefix length: 32

Oversampling factor: 1

Number of OFDM symbols: 5

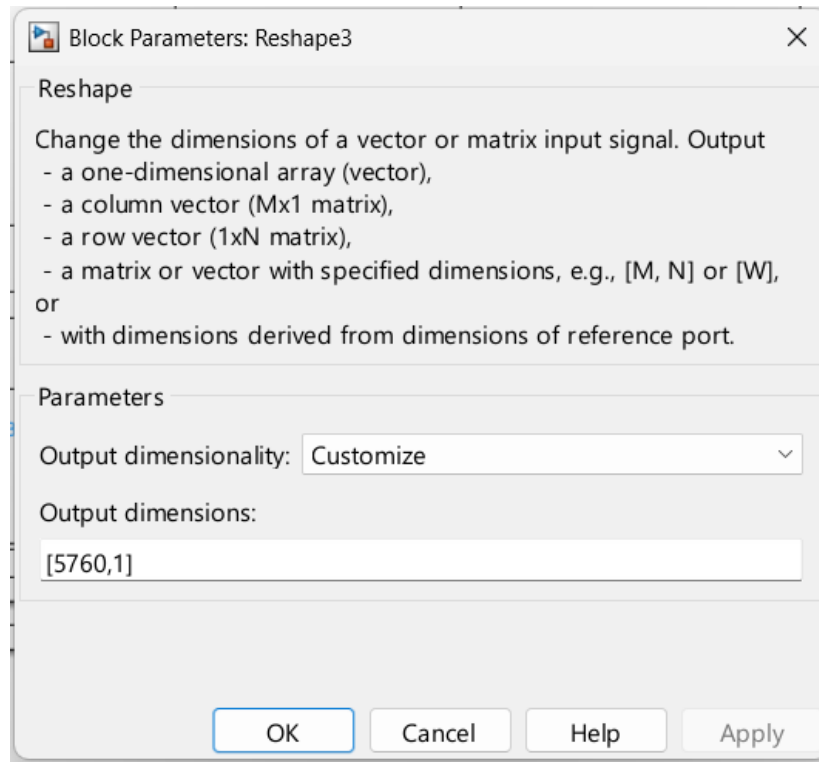
Number of receive antennas: 1

Simulate using: Interpreted execution

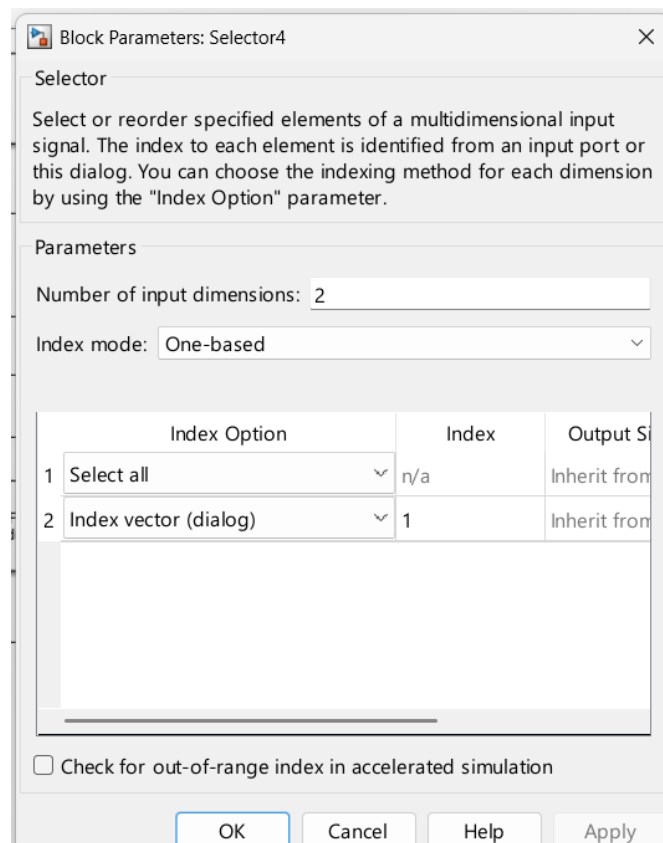
OK Cancel Help Apply

MIMO Gain branch

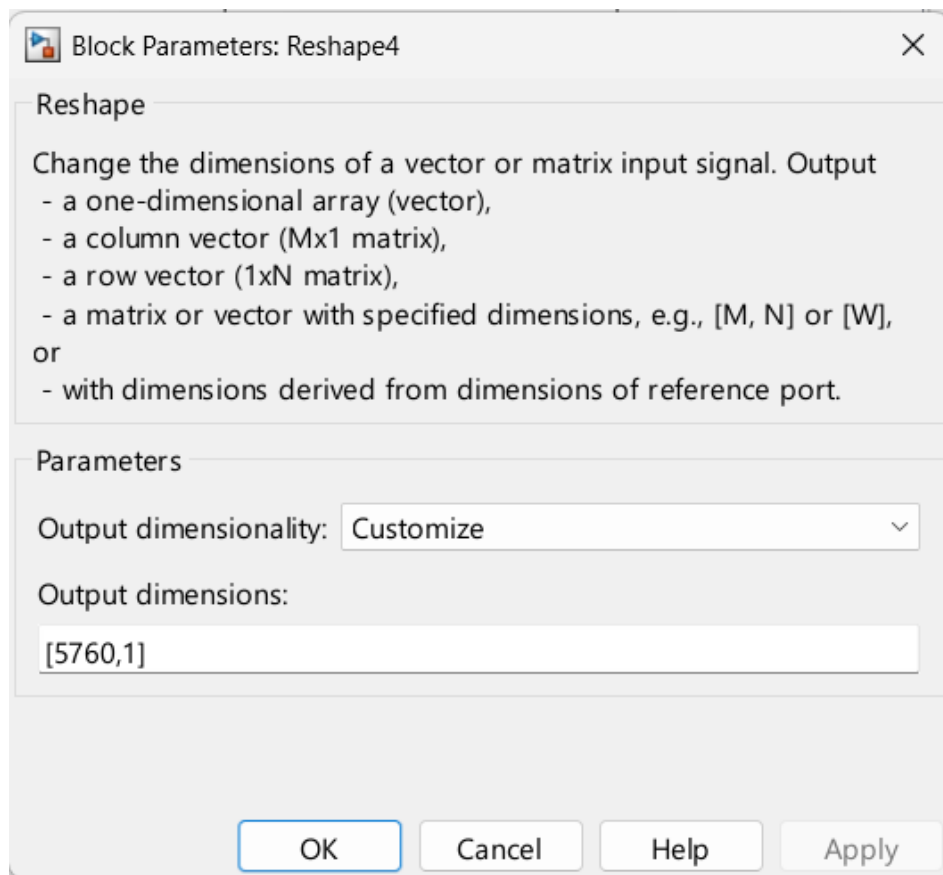
Reshape after MIMO Gain output



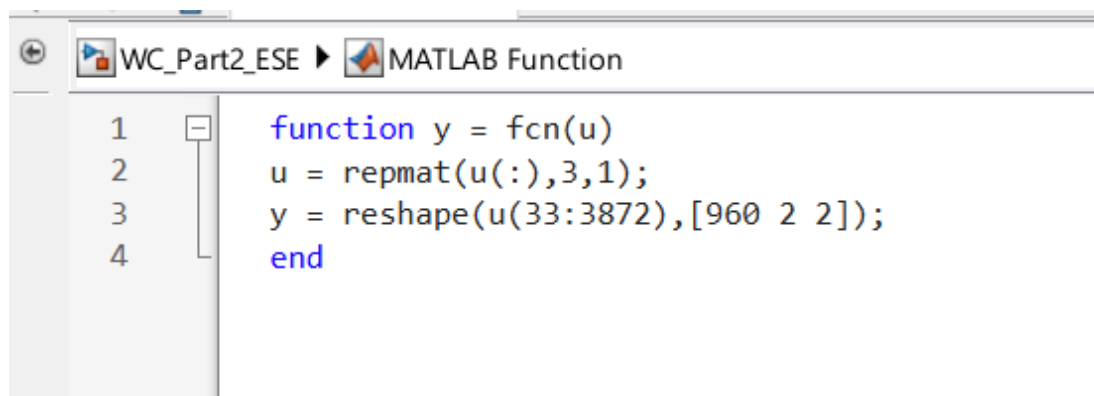
Selector after Reshape(GAIN Branch)



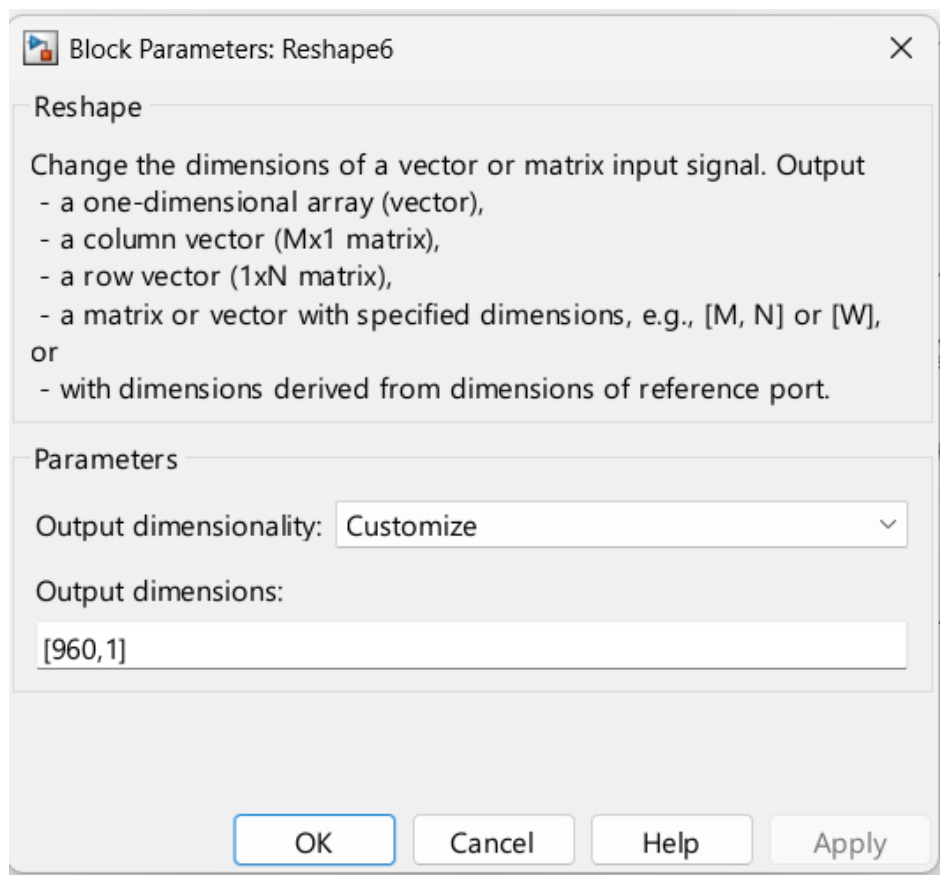
Reshape after above selector block



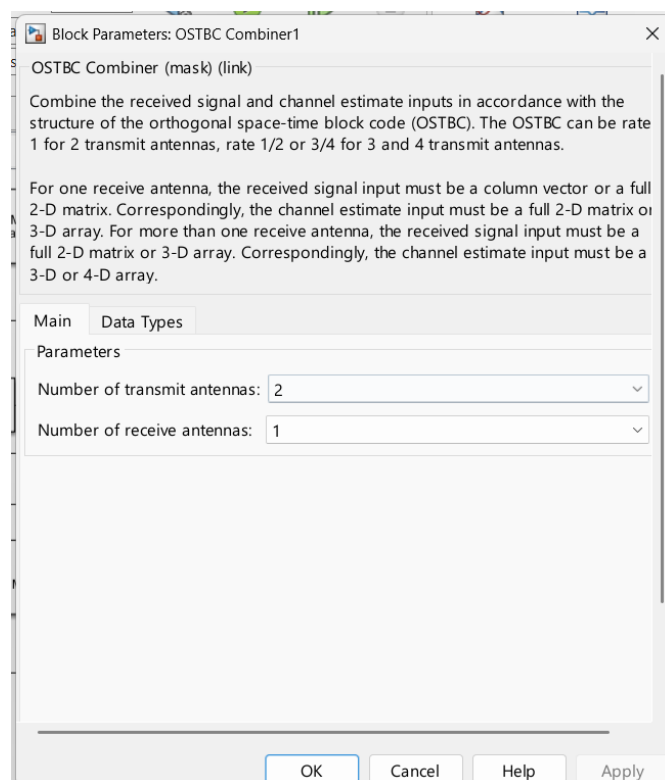
Function Block(Expand Channel) after Reshape



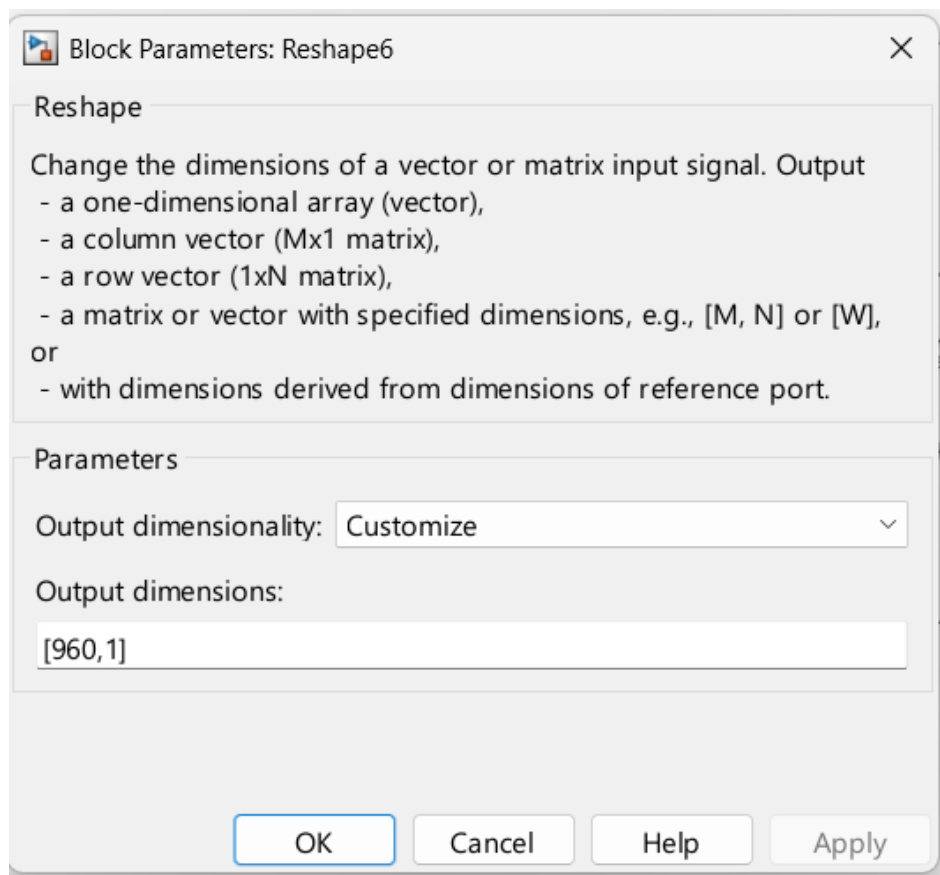
Reshape after Function Block



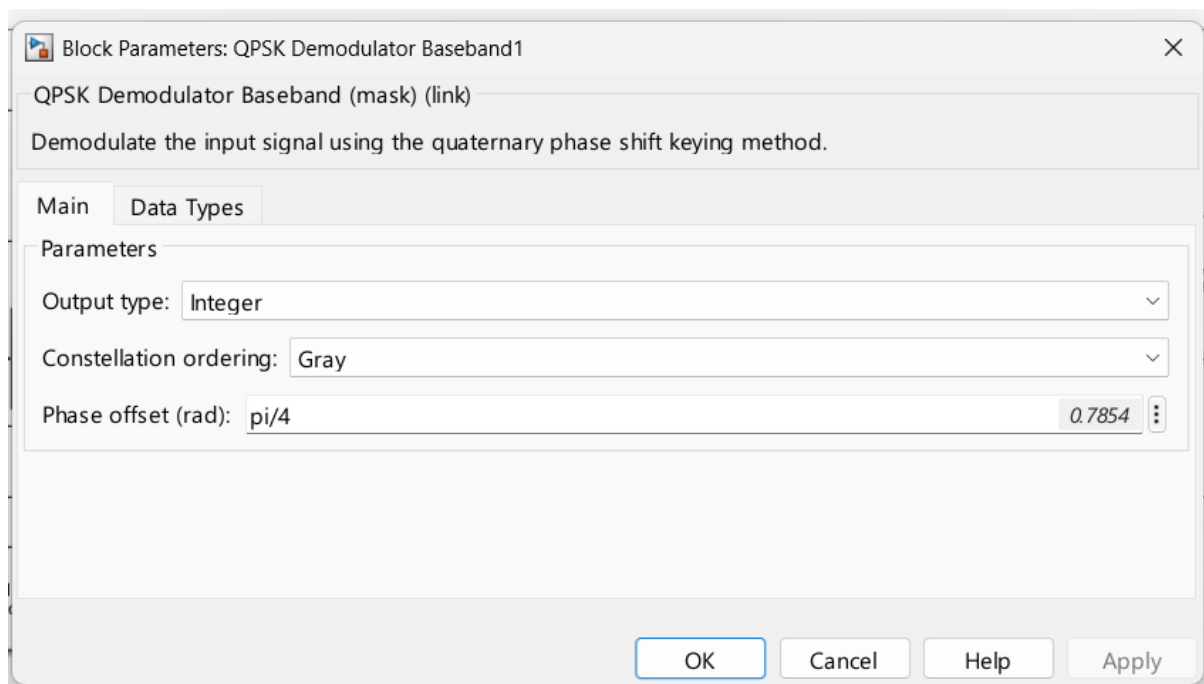
OSTBC Combiner



Reshape after OSTBC Combiner



QPSK Demodulator



Error Rate Calculation

 Block Parameters: Error Rate Calculation ✕

Error Rate Calculation (mask) (link)

Compute the error rate of the received data by comparing it to a delayed version of the transmitted data. The block output is a three element vector consisting of the error rate, followed by the number of errors detected and the total number of symbols compared. This vector can be sent to either the workspace or an output port.

The delays are specified in number of samples, regardless of whether the input is a scalar or a vector. The inputs to the 'Tx' and 'Rx' ports must be scalars or column vectors.

The 'Stop simulation' option stops the simulation upon detecting a target number of errors or a maximum number of symbols, whichever comes first.

Parameters

Receive delay:

0

⋮

Computation delay:

0

⋮

Computation mode:

Entire frame

▼

Output data:

Port

▼

☐ Reset port

OK

Cancel

Help

Apply

EVM Measurement

 Block Parameters: EVM Measurement ✕

EVM Measurement

Calculate the RMS Error Vector Magnitude (EVM) between a reference signal and a corrupted received signal of an input frame. The RMS EVM measurement is defined in different standards as the (%) ratio of the RMS error to the square root of either one of the following:

- 1) Average reference signal power
- 2) Average constellation power
- 3) Peak constellation power

All power values are given in Watts, referenced to 1 ohm.

The optional maximum EVM is the maximum EVM value in an input frame. The optional X-percentile output is the EVM value below which X% of the computed EVM values lie.

[Source code](#)

Parameters

Normalize RMS error vector by:	Average reference signal power
Reference signal:	Input port
Measurement interval:	Input length
Averaging dimensions:	1
☒ Output maximum EVM	
☐ Output X-percentile EVM	

Simulate using:

Interpreted execution

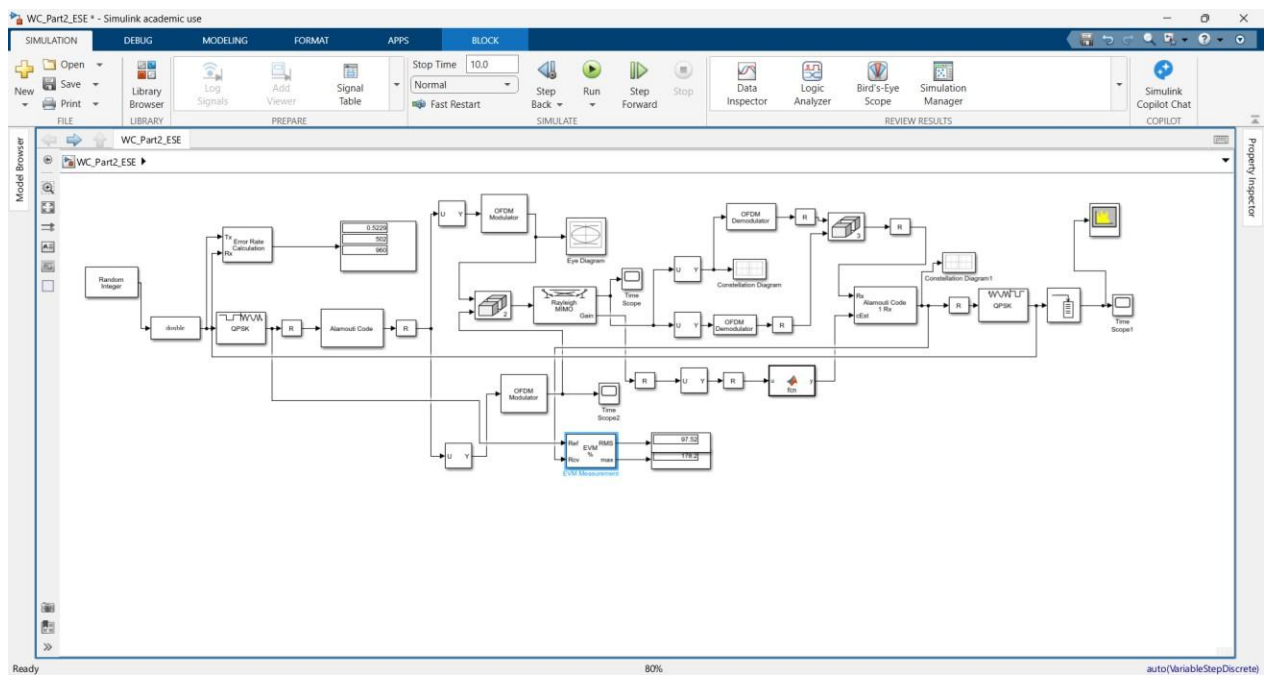
OK

Cancel

Help

Apply

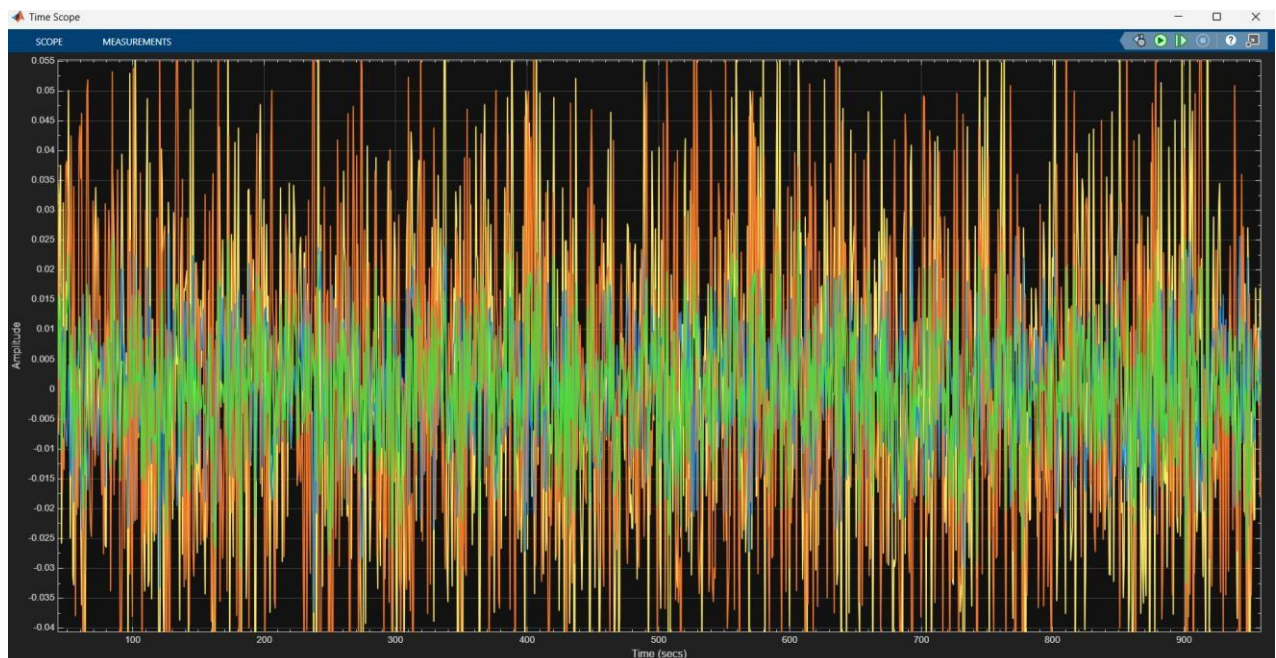
(e). Total screen shot of working Simulink model.



Simulink model

Results

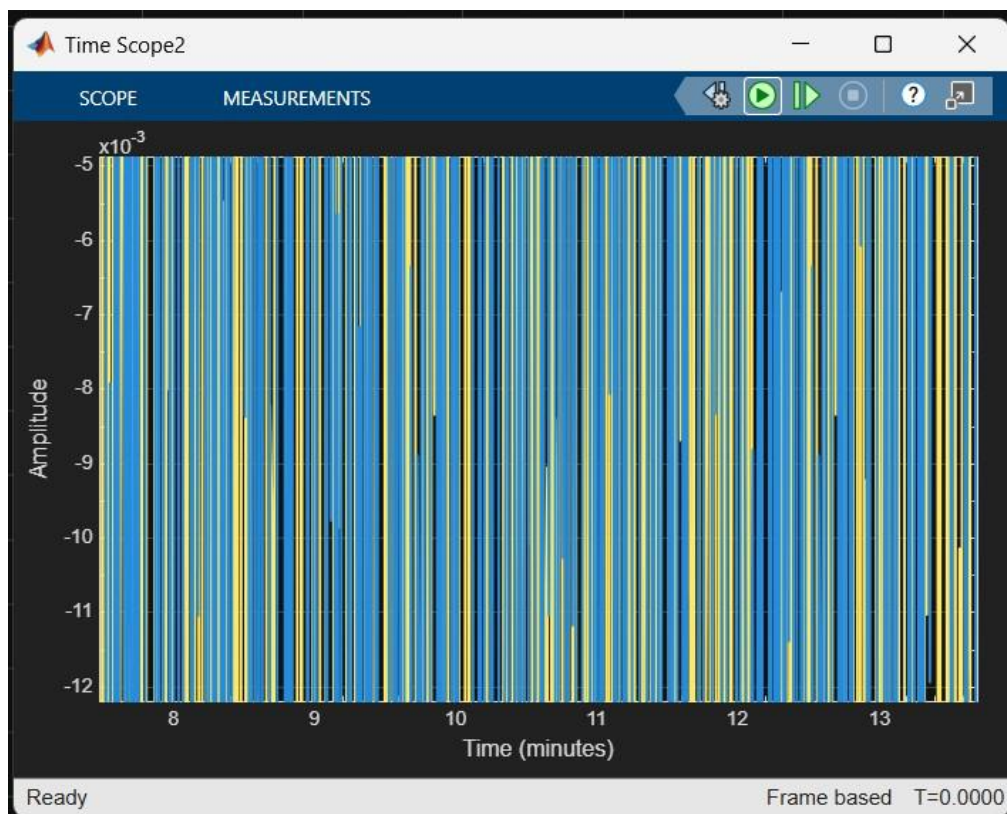
(c) Screen shots of time domain original and decoded signal



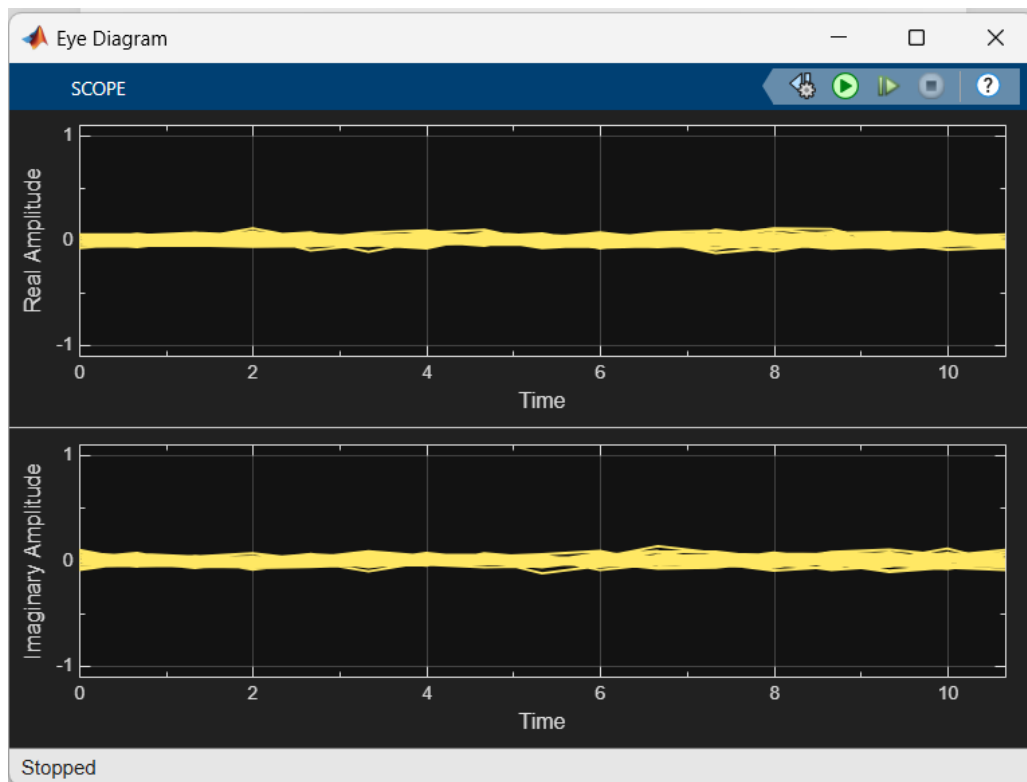
Time Scope(MIMO)



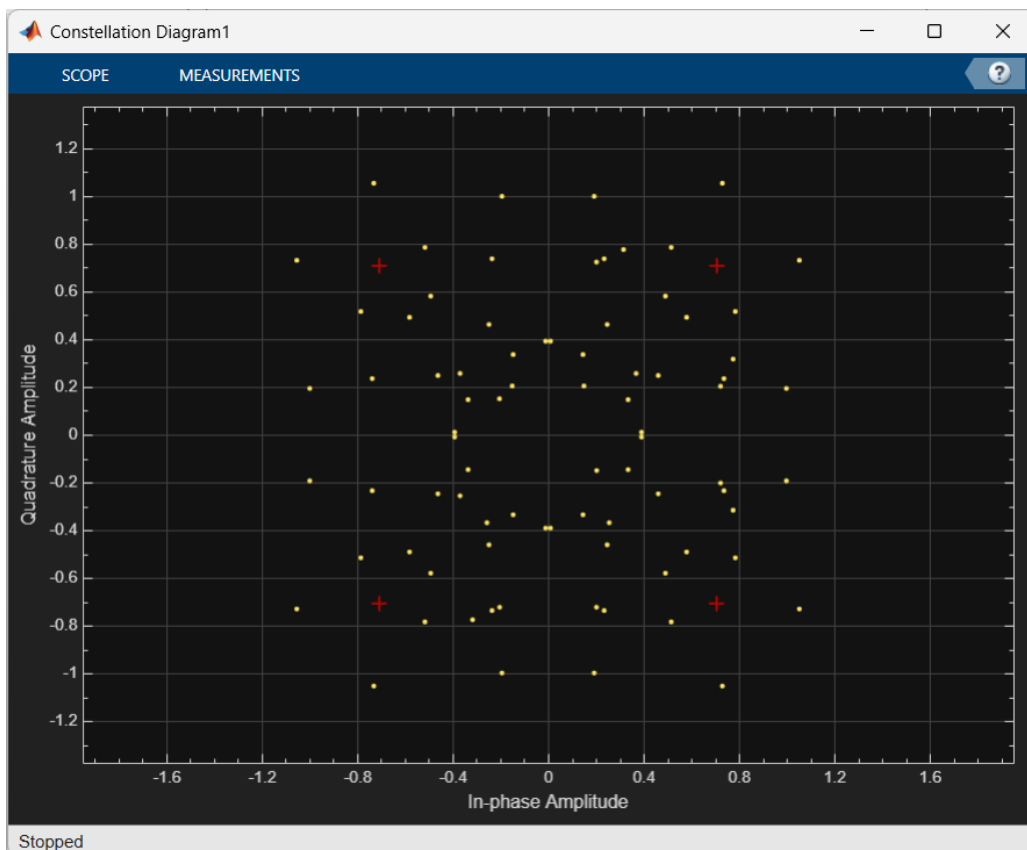
Time Scope(QPSK)



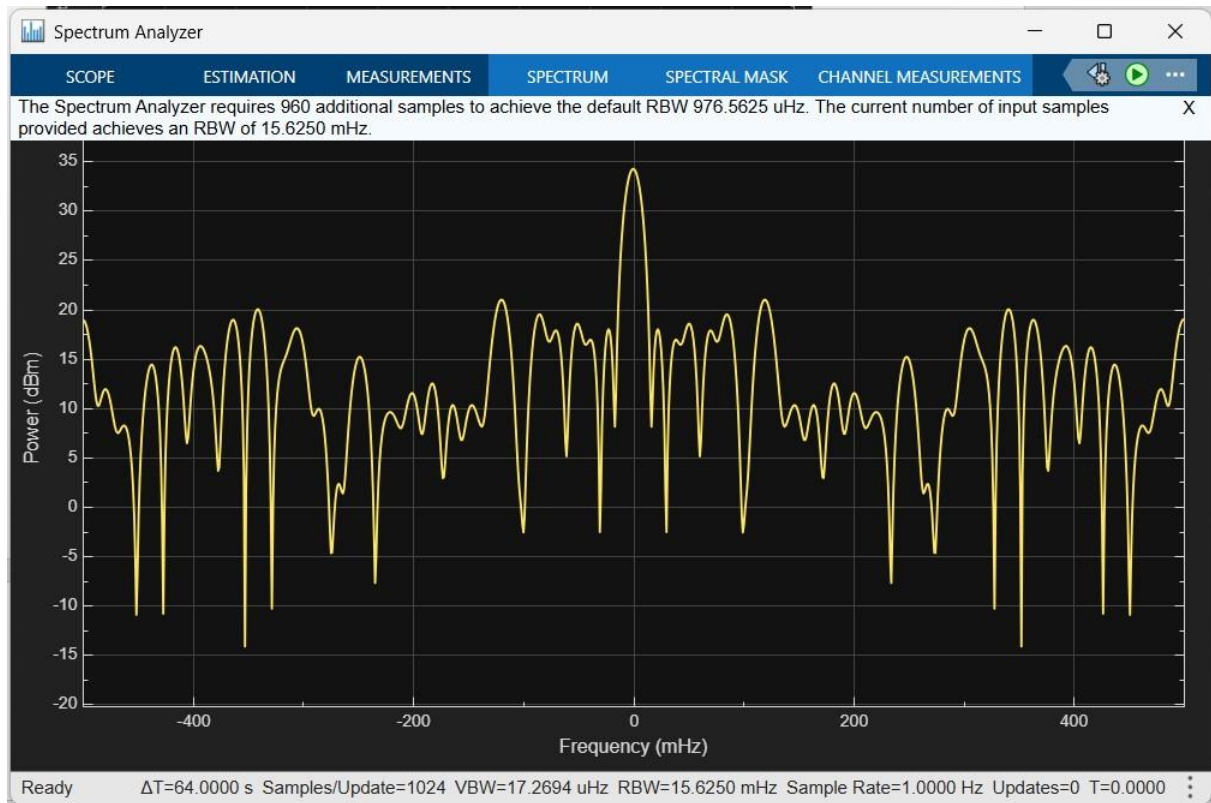
Time Scope



Eye Diagram



Constellation Diagram



Spectrum Analyzer

